EXPERIMENT 4

AIM

To create a simple web application that includes a form to accept a user's name and age, and uses a Servlet to process the submitted data and display the values in the browser.

ALGORITHM

1. Input:

- *The user enters their name and age in the web form and submits it.*

2. Form Handling and Servlet Processing:

1. Create the Web Form:

- *Design an HTML form with input fields for name and age.*
- *Add a submit button.*
- *Set the form action to the Servlet URL and method to POST.*

2. Create the Servlet Class:

- *Develop a Servlet class extending HttpServlet.*
- *Override the doPost() method to handle form submission.*
- *Retrieve form data using request.getParameter("username") and request.getParameter("age").*

3. Process the Data in the Servlet:

- *Extract the entered name and age values.*
- *Perform basic validation (e.g., check if fields are not empty).*
- *Prepare a response message using the input data.*
-

4. Send Response to the Browser:

- *Set response content type as text/html.*
- *Use PrintWriter to display output like:*
  "Hello [name], You are [age] years old."

5. Deploy the Servlet:

- *Compile the Servlet program.*
- *Deploy it in a Servlet container like Apache Tomcat.*
- *Ensure proper URL mapping using annotations or web.xml.*

6. Test the Application:

- *Open the browser and load the HTML form.*
- *Enter details and submit the form.*
- *View the processed result displayed on the browser.*

CODE:

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>Passenger Details</title>

<style>
body{
    font-family: Arial;
    background: linear-gradient(135deg,#1e3c72,#2a5298);
    margin:0;
}

header{
    background:#0d47a1;
    color:white;
    text-align:center;
    padding:15px;
    font-size:22px;
}

.container{
    width: 420px;
    margin: 80px auto;
    padding: 25px;
    background: rgba(255,255,255,0.15);
    backdrop-filter: blur(10px);
    border-radius: 12px;
    box-shadow: 0px 0px 15px rgba(0,0,0,0.3);
    color:white;
}
```

```
h2{
    text-align:center;
    margin-bottom:20px;
}

.form-group{
    display:flex;
    align-items:center;
    margin-bottom:15px;
}

label{
    width:100px;
}

input{
    flex:1;
    padding:8px;
    border-radius:5px;
    border:none;
    outline:none;
}

input:focus{
    box-shadow:0 0 6px #00e5ff;
}
```

```
.submit-btn{
    text-align:center;
    margin-top:15px;
}

input[type="submit"]{
    background: linear-gradient(45deg,#00e5ff,#00c853);
    color:white;
    border:none;
    padding:10px 20px;
    border-radius:25px;
    cursor:pointer;
}

input[type="submit"]:hover{
    transform:scale(1.05);
}
</style>

</head>

<body>

<header>
    Online Bus Pass Application System
</header>
```

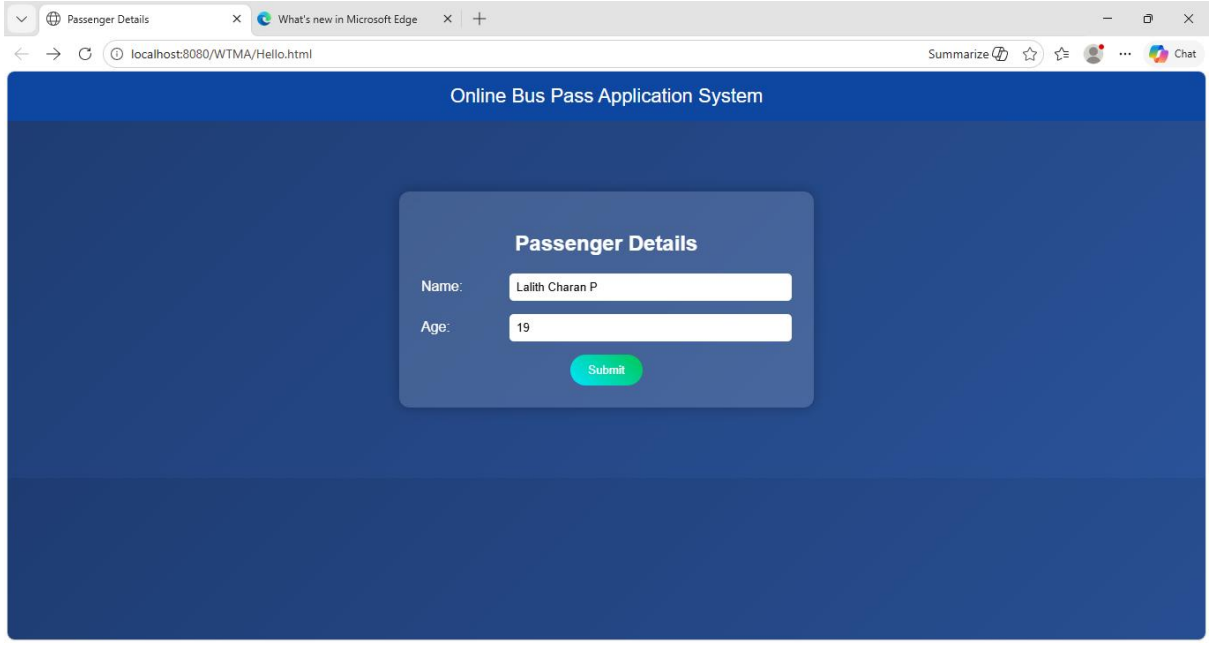
```
<div class="container">
<h2>Passenger Details</h2>

<form action="UserServlet" method="post">

<div class="form-group">
<label>Name:</label>
<input type="text" name="username" required>
</div>

<div class="form-group">
<label>Age:</label>
<input type="number" name="age" required>
</div>

<div class="submit-btn">
<input type="submit" value="Submit">
</div>

</form>
</div>

</body>
</html>
package WTMA;
```

```
import jakarta.servlet.ServletException;
import jakarta.servlet.annotation.WebServlet;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import java.io.IOException;
import java.io.PrintWriter;

@WebServlet("/UserServlet")
public class UserServlet extends HttpServlet {

    private static final long serialVersionUID = 1L;

    protected void doPost(HttpServletRequest request,
HttpServletResponse response)
            throws ServletException, IOException {

        response.setContentType("text/html");

        PrintWriter pw = response.getWriter();

        String name = request.getParameter("username");
        String age = request.getParameter("age");

        pw.println("<!DOCTYPE html>");
        pw.println("<html><head><title>Result</title>");

        // CSS
```

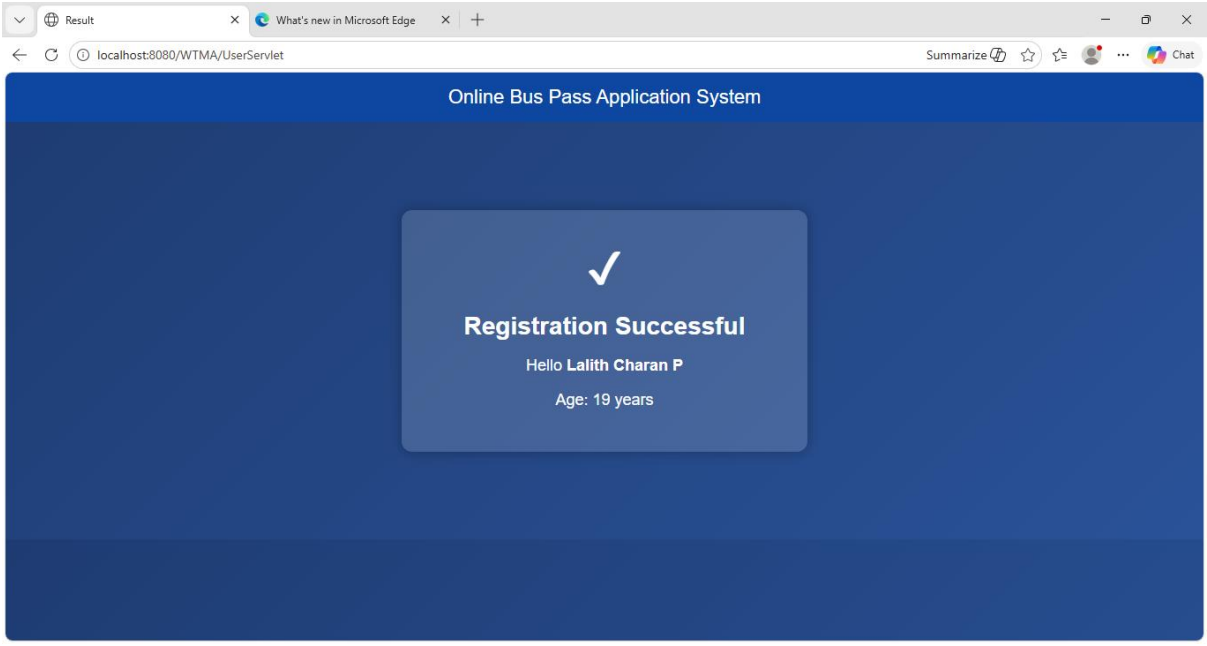
```
        pw.println("<style>");
        pw.println("body{margin:0;font-family:Arial;background:linear-gradient(135deg,#1e3c72,#2a5298);color:white;text-align:center;}");
        pw.println("header{background:#0d47a1;padding:15px;font-size:22px;}");
        pw.println(".card{margin:100px auto;width:400px;padding:30px;border-radius:12px;background:rgba(255,255,255,0.15);box-shadow:0 0 15px rgba(0,0,0,0.3);}");
        pw.println("h1{font-size:28px;margin-bottom:10px;}");
        pw.println("p{font-size:18px;}");
        pw.println(".icon{font-size:50px;margin-bottom:10px;}"); // simple graphic
        pw.println("</style>");


        pw.println("</head><body>");


        pw.println("<header>Online Bus Pass Application System</header>");


        pw.println("<div class='card'>");
        pw.println("<div class='icon'>✓</div>"); // graphic (not bus)
        pw.println("<h1>Registration Successful</h1>");
        pw.println("<p>Hello <b>" + name + "</b></p>");
        pw.println("<p>Age: " + age + " years</p>");
        pw.println("</div>");


        pw.println("</body></html>");
    }
```

}

OUTPUT 1:



OUTPUT 2:



RESULT

The web application successfully accepted user input through a form, processed it using a Servlet, and displayed the entered name and age in the browser.